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Fifth-generation Drones

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6.1 Introduction

The fifth generation of drones marks the dawn of a new era in unmanned aerial technology, an age where complete autonomy, cognitive artificial intelligence (AI), quantum communication, and human–machine symbiosis converge to redefine aerial operations. These drones transcend the limitations of earlier generations by exhibiting real-time self-learning, adaptive decision-making, and fully cooperative mission execution without human intervention. Built upon the technological foundation laid by fourth-generation systems, fifth-generation UAVs integrate advanced capabilities such as swarm intelligence 2.0, edge computing, neural-network-based flight control, quantum-encrypted communication, and interconnected multi-domain warfare systems (Chattopadhyay, 2024).

At their core, fifth-generation drones are not merely aerial machines; they are autonomous cognitive entities capable of perceiving, analyzing, and acting independently within dynamic and unpredictable environments. Unlike their predecessors, which relied heavily on human supervision or centralized control, these drones utilize a distributed AI architecture that enables collective learning, inter-drone coordination, and mission adaptability across air, land, sea, and space domains. This allows them to perform simultaneous operations such as target identification, real-time